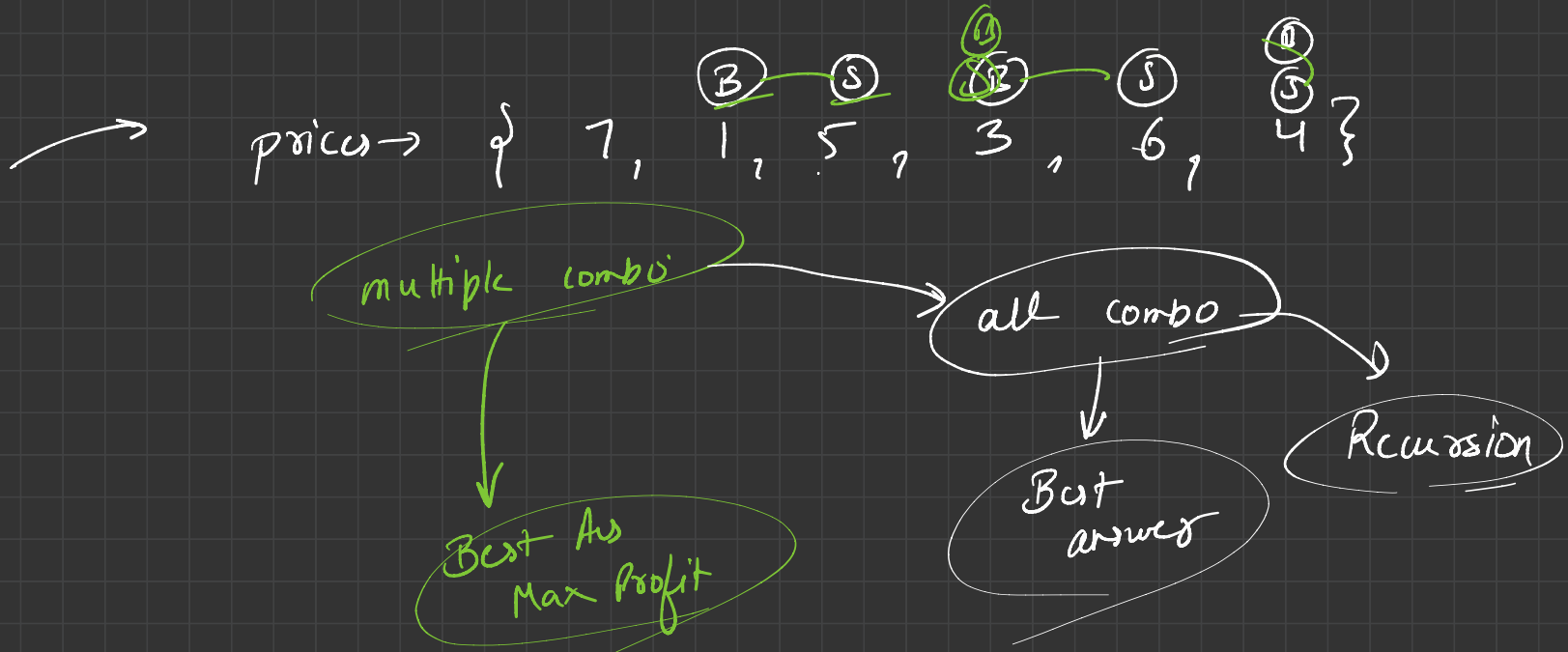
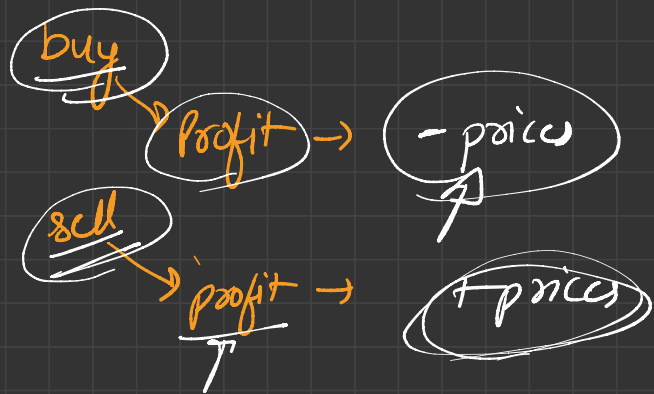
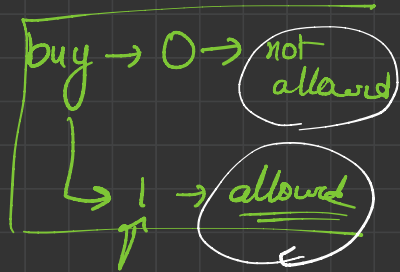
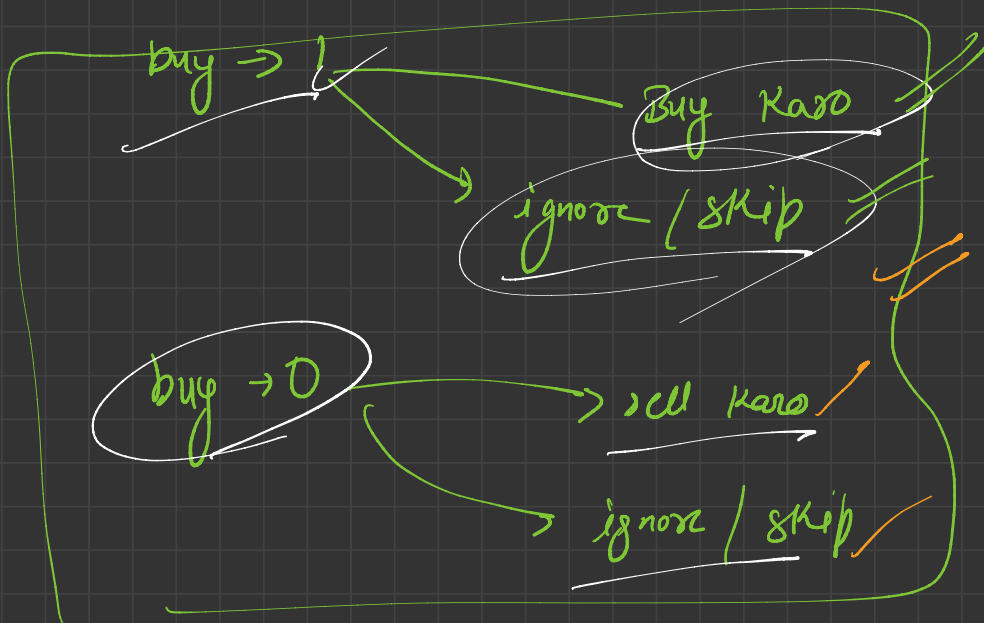
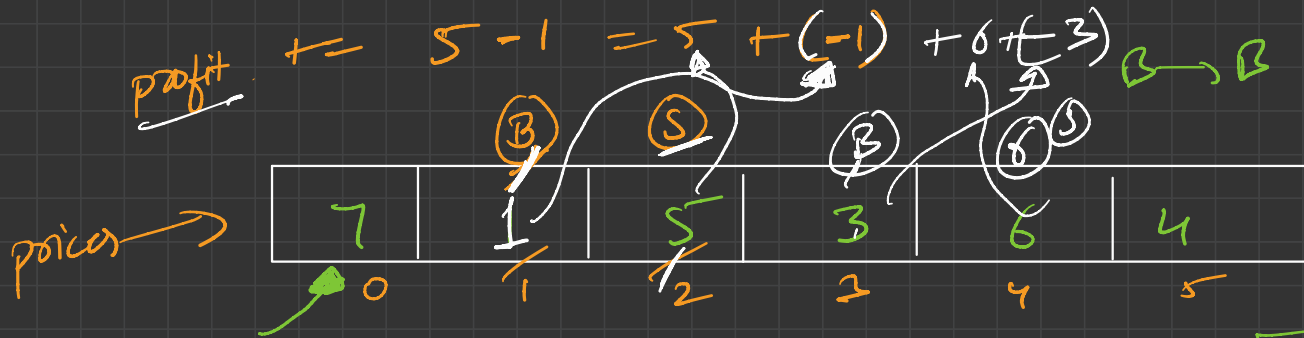



Dynamic Programming





price →

7	1	5	3	6	4
---	---	---	---	---	---

0 1 2 3 4 5

B
↑
i
3

logic

if (buy)

$$\text{profit} = \boxed{-\text{price}[i] + f(i+1, 0);}$$

index buy

$$\boxed{0 + f(i+1, 1);}$$

skip

buy
karo

max

else
{

// buy $\rightarrow 0$

$$\text{profit} = \begin{array}{l} \boxed{+ \text{price}[i] + f(i+1, 1)} \\ \text{or} \\ \boxed{0 + f(i+1, 0)} \end{array}$$

sell karo

max

skip karo

Rec

→ + Mem

① dp array create
func calls add

② ans return
↳ dp me store

③ B.C → K back
dp check Kalo → answer already present
↙
return Kalo

Top - Down

index $\rightarrow$
 $0 \rightarrow n$

Bottom - Up

$n-1 \rightarrow 0$

S.O $\rightarrow ?$ $O(n)$ $\rightarrow O(1) =$

dp[index][buy]

